

BACCHIRI! — Verifiable Measurement Layer

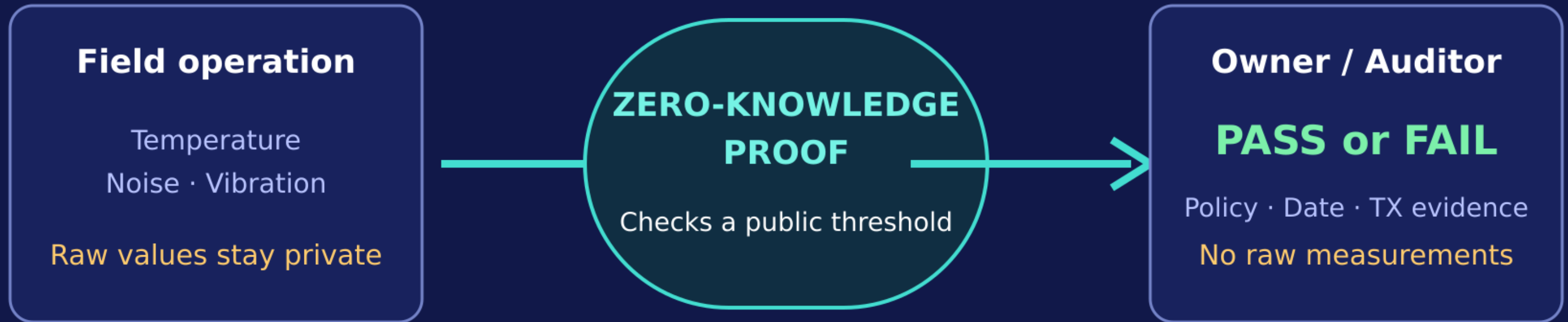
**Prove threshold compliance
without revealing sensor values**

Midnight × Cloudflare × Edge Device

English demo pitch

THE TRUST PROBLEM

Share compliance — not raw sensor data



Contractors, equipment providers, owners, and auditors can verify the same result.

One controlled path from policy to public evidence

The public threshold is fixed before private measurements become a proof.



PUBLIC BEFORE OPERATION

MINIMUM PUBLIC EVIDENCE

Bind the Device to a public threshold before measurement

Connected wallet, Device identity, registered policy, and validity period.

Location: Device Workflow

Device Workflow

Register this PC as a Device, capture a sensor value, create a ZK proof, and record the claim on Midnight.

PREPROD

INFO

Stored Device Identity restored

Project

Select Project

test1

+ New Project

Projects per Wallet: 2 / 10

Policy

+ New Policy

Policies in this Project: 1 / 10

Policy	Public threshold	Status	Policy registration TX
test policy-abfd6...ae3fc618ab9f	Between minimum and maximum 20.5-35 ° C	Registered	00fcbde2c825...42fe0680eed ?

Use-Case Progress

1 Device registration

2 Threshold assignment

3 Sensor value

4 ZK proof

5 Midnight record

1. Device Identity

Device ID (derived from Wallet)
device-a9c4e7eba069c0d01f337da55d7b1f915a819b3c3dceb8c67053421b364b41fc

✓ Create Device Identity

Wallet
lace / mn_shield-add...3pj6ruqc5e61e

Device Authority
ee24a9d4dec6766...7a8d910830687d8

2. Midnight registration

Policy
test / 20.5-35 ° C

20.5-35 °C
00fcbde2c825296894794df8df79443bdd74e3fb1b0ee1d16b96d642fe0680eed

Register Device and assign policy

NEXT ACTION

Contract: 10cb9e430180afca0...bc236b521eac0b2f2

3. Sensor value

Sensor date (JST)

Previous day

2026/08/26

Next day

Generation mode

Within threshold (ZKP test)

Previous 30 completed days only: 2026-08-01 - 2026-08-30

- 1 User-controlled Wallet
- 2 Device identity
- 3 Public threshold
- 4 Validity period

PRIVATE SENSOR EVIDENCE

24 ordered hourly slots become the private witness

Edge Device

Individual readings
Hourly minimum / maximum

PRIVATE



A day is represented explicitly

00:00
MIN / MAX

01:00
MIN / MAX

02:00
STOPPED

03:00
MIN / MAX

... 24

Missing data is STOPPED — never silently PASS

Only the proof result is shared outside the device.

Generate the ZK proof against the registered policy

Private hourly extrema form the witness; the threshold cannot be replaced at proof time.

Device registration TX0041bb31a5b461c78a6dd0ddf260a76405f246cbe684acbf716f13c1ebd38d03

Threshold assignment TX00c15e249991c7ebbb9261a1bae8b11a105a797467430f16e8c88e120ae77d3831

Contract: 10cb9e430180afca0..bc236b521eac0b2f2

3. Sensor value

Sensor date (JST)2026/08/28

Generation modeWithin threshold (ZKP test)

Previous 30 completed days only: 2026-08-01 - 2026-08-30

Auto Generate one day

Period2026-08-28

Sensor readings used1440

Outliers0

Threshold resultWITHIN THRESHOLD

Attestation164352c4efb6136..b2e27145f8fc95f

4. ZK proof

Request proof processingIN PROGRESS

Proof Job-

Status-

5. Midnight record

Generate proof and record TX

Processing status: Request proof processing

TX: -

TRANSACTION FEE SPONSORED

The service Sponsor Wallet adds the DUST fee and submits the approved Device transaction.

Daily sponsored submissions (JST)

20 remaining / 20 (0 used)

Resets: 01/09/2026, 00:00:00

Daily sensor history

PRIVATE INPUT

Hourly MIN / MAX

STOPPED slots

Proof nonce

TO

PUBLIC OUTPUT

WITHIN / OUTSIDE

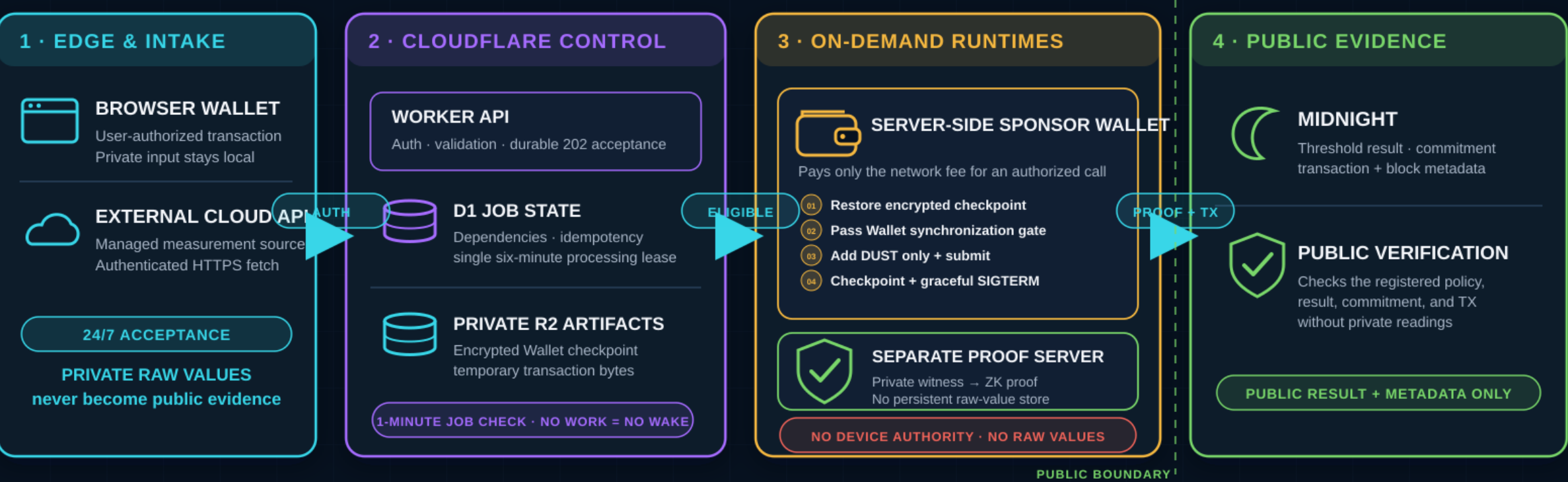
Observed hours

Commitment

BACCHIRI! · MIDNIGHT BUILDATHON

ON-DEMAND ZK PROOF & MIDNIGHT RECORDING

24/7 admission with a stateful Sponsor Wallet that wakes only for eligible work, pays DUST, checkpoints, and stops.



COST-AWARE OPERATIONS · STANDARD-4 PLANNING EXAMPLE

\$133.23
24/7 · 720 h / 30 days
Excludes Workers Paid, D1, Queue, R2, egress, and Proof Server; actual cost follows drain duration.

➔

\$22.20
illustrative 4 h/day drain · 120 h

-83.3%
PLANNING ESTIMATE

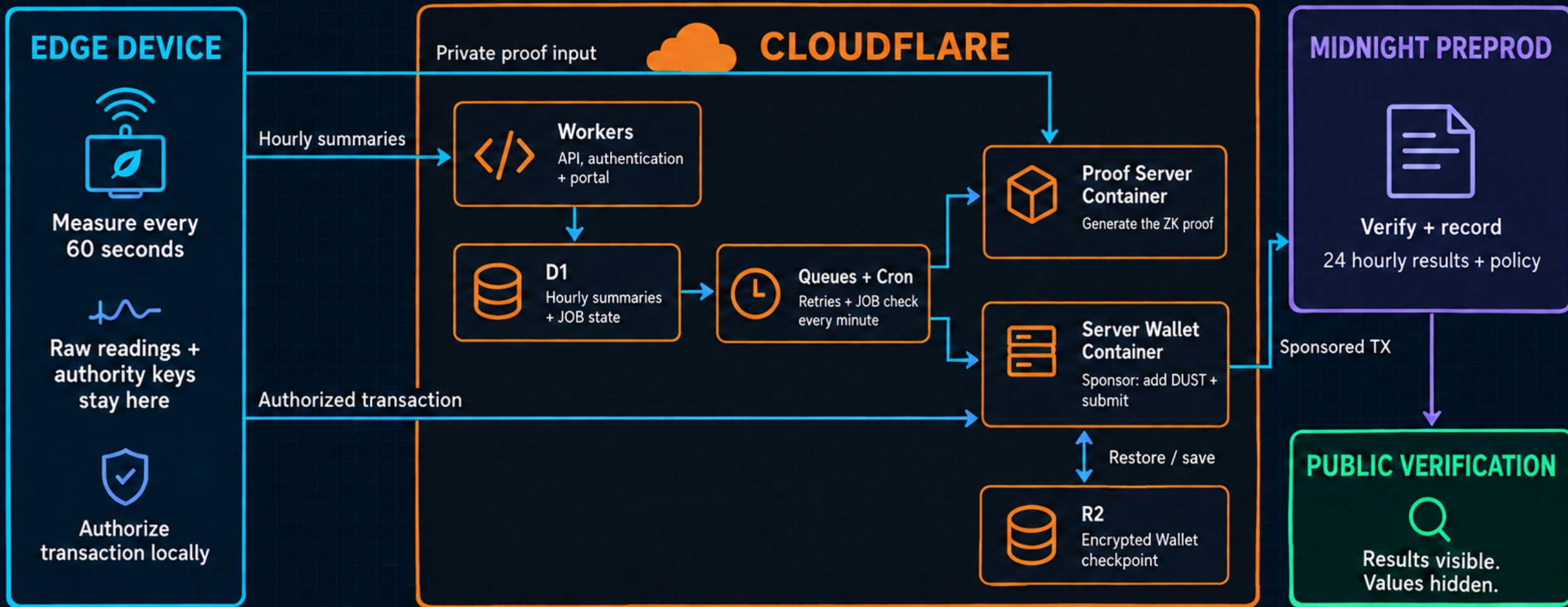
SERVER-WALLET ACHIEVEMENT

BUILT + VALIDATED

System pays the DUST fee for an authorized call
Device authority and private values stay separate
Preprod flow confirmed · 2026-09-03

CONTINUOUS MEASUREMENT. SCHEDULED PROOF.

What runs on Cloudflare



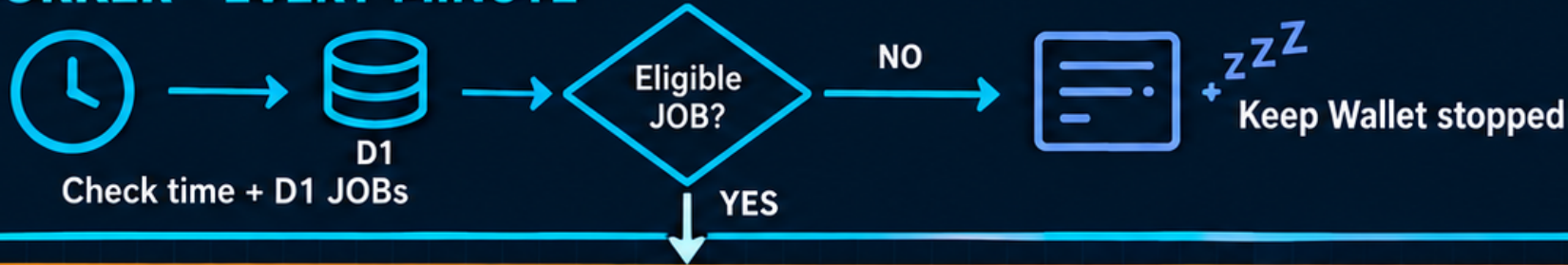
🕒 **02:00 JST • START PROCESSING THE PREVIOUS DAY**

Trusted backend receives private proof input. Sensor values are hidden from public viewers.

THE WALLET RUNS ONLY WHEN WORK IS DUE

Scheduled mode • daily processing starts at 02:00 JST

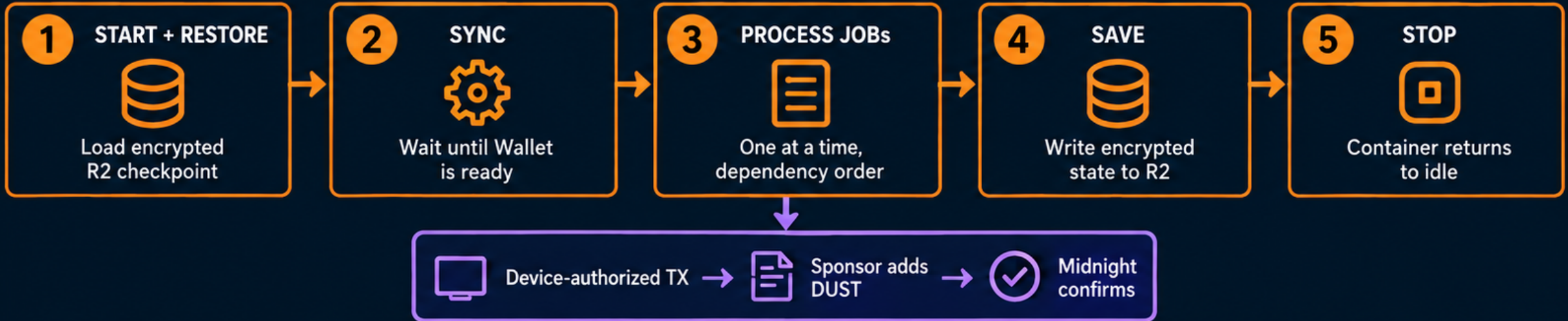
WORKER • EVERY MINUTE



Next eligible batch: check again

New JOBS after the daily cutoff wait for the next run.

SERVER WALLET CONTAINER • ONLY WHILE PROCESSING



SENSOR COLLECTION CONTINUES EVERY 60 SECONDS

02:00 is the processing start. Completion follows synchronization, proof generation and confirmation.

A third party checks public evidence — not private measurements

28/08/2026, 01:00:00 28/08/2026, 02:00:00	60	21.49 °C	22.39 °C	21.94 °C	6f0a70803..27f2c901b
28/08/2026, 02:00:00 28/08/2026, 03:00:00	60	22.37 °C	23.21 °C	22.78 °C	2129fab02..49345391d
28/08/2026, 03:00:00 28/08/2026, 04:00:00	60	23.16 °C	23.82 °C	23.49 °C	e6baef1b29..dffe9cod5
28/08/2026, 04:00:00 28/08/2026, 05:00:00	60	23.73 °C	24.18 °C	23.98 °C	1568dad39..a05cb1513
28/08/2026, 05:00:00 28/08/2026, 06:00:00	60	24.1 °C	24.31 °C	24.22 °C	ba9cb1acf..56dc75a95
28/08/2026, 06:00:00 28/08/2026, 07:00:00	60	24.07 °C	24.31 °C	24.22 °C	219e55eba..0df505f1e
28/08/2026, 07:00:00 28/08/2026, 08:00:00	60	23.73 °C	24.16 °C	23.97 °C	add55e8de..672559262
28/08/2026, 08:00:00 28/08/2026, 09:00:00	60	23.15 °C	23.82 °C	23.49 °C	222fed10d..33b29fa59
28/08/2026, 09:00:00 28/08/2026, 10:00:00	60	22.38 °C	23.22 °C	22.8 °C	34e2db9cc..cabb87ac5
28/08/2026, 10:00:00 28/08/2026, 11:00:00	60	21.49 °C	22.45 °C	21.96 °C	cbe3819e4..bb50479cd
28/08/2026, 11:00:00 28/08/2026, 12:00:00	60	20.5 °C	21.54 °C	21.01 °C	9b8dbe29c..f77ca9af4
28/08/2026, 12:00:00 28/08/2026, 13:00:00	60	19.57 °C	20.57 °C	20.03 °C	e1ac863f6..5af19f570
28/08/2026, 13:00:00 28/08/2026, 14:00:00	60	18.69 °C	19.57 °C	19.09 °C	277fe467a..d9a3cd42e
28/08/2026, 14:00:00 28/08/2026, 15:00:00	60	17.85 °C	18.68 °C	18.24 °C	a54af9db9..a6942c649
28/08/2026, 15:00:00 28/08/2026, 16:00:00	60	17.2 °C	17.87 °C	17.54 °C	aa03b5b17..eee489f5a
28/08/2026, 16:00:00 28/08/2026, 17:00:00	60	16.84 °C	17.31 °C	17.05 °C	ec844aa2f..3c217d159
28/08/2026, 17:00:00 28/08/2026, 18:00:00	60	16.71 °C	16.93 °C	16.8 °C	f2d6570cc..0704805f0
28/08/2026, 18:00:00 28/08/2026, 19:00:00	60	16.71 °C	16.94 °C	16.79 °C	0ac4d6ade..3ff25d5cd
28/08/2026, 19:00:00 28/08/2026, 20:00:00	60	16.83 °C	17.29 °C	17.03 °C	39bf993e6..597c4e2f2
28/08/2026, 20:00:00 28/08/2026, 21:00:00	60	17.24 °C	17.86 °C	17.53 °C	d2a74981c..c62d8eed7
28/08/2026, 21:00:00 28/08/2026, 22:00:00	60	17.83 °C	18.66 °C	18.22 °C	584f2aa4d..c1ef2c53e
28/08/2026, 22:00:00 28/08/2026, 23:00:00	60	18.6 °C	19.54 °C	19.07 °C	0b80e3111..f46bc8c39
28/08/2026, 23:00:00 29/08/2026, 00:00:00	60	19.48 °C	20.5 °C	20.01 °C	561539821..5522f018f

Data source: Device / Cloudflare / Midnight

Sensor values stay private; anyone can check the threshold result on Midnight

31/08/2026, 11:37:19

Public Result

Every submitted hourly minimum and maximum for the observed hours was within the registered public threshold policy; hours without data are reported as STOPPED.

Information Anyone Can Check

Proof record ID	proof-6a78a68e7df6a8e26080bfafeb76b61ca8e7ff11e792c85
Period	2026-08-28
Sensor readings used	1440
Observed hours	24 / 24
STOPPED hours	0 / 24
Threshold result	WITHIN THRESHOLD
Threshold setting ID	policy-0ac49603-0b72-4924-b650-9cd233f8d1 b2
Threshold rule	Between minimum and maximum
Public threshold	15.51-25.51 ° C
Policy version	1
Sensor	Temperature / ° C
Applied policy record	f54710ae0df6b0a4a364a04846599f371d563600671f30c185ecd2c870881a1d3
Schema version	5
Circuit version	3
Data fingerprint	164352c4e2b6136a58e3d0be6bbb7462ff0e8787ee1c06c981b2e2714828fc95f
Status	Recorded on Midnight
ZKP generated at	-
Network	Midnight Preprod
Contract	10cb9e430180afca0c6840a16dc4a20e9cefa53f27d4aa5bc236b521eac0b2f2..?
Attestation Transaction	0093ae437b051d0ec0942c85e458f4a9a313be7dcd5958c7d422899030456262bb
Transaction hash	279caa57ba992cffe7b9255bb170f8196666f13b32eae262c1c9f4d46763879a..?
Block height	2338540..?

What Was Checked

CHECKED

Daily proof record found

CHECKED

Hidden hourly summary is tied to this proof

CHECKED

ZK proof matches the published threshold result

CHECKED

Midnight transaction confirmed

Information Hidden from Third Parties

RAW SENSOR VALUES ... HIDDEN

HOURLY MINIMUM / MAXIMUM HIDDEN

RANDOM PROOF SECRET HIDDEN

OBSERVED / STOPPED HOURS PUBLIC ON MIDNIGHT

THRESHOLD RULE PUBLIC ON MIDNIGHT

PUBLIC

Day · Device · Threshold · Observed / STOPPED · Result · Commitment · Transaction

PRIVATE

Readings · Hourly MIN / MAX · Nonce

AN HONEST CLAIM BOUNDARY

What this proof establishes — and what it does not

PROVES

- ✓ Submitted private hourly extrema
- ✓ Checked against the registered policy
- ✓ Bound to public result and TX evidence

DOES NOT PROVE

- × Physical sensor accuracy
- × Complete real-world sampling
- × Correct aggregation on the device

A precise claim is evidence that third parties can use responsibly.